PILOT: an offline routing study

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Complete local scope: 8 evidence-admitted model arms, 3 replicate seeds, 13 primary methods, 8 budget fractions, 4 learning prefixes and 5 controlled ablations. All 1,098 deployment conditions and 108 method/variant groups passed independent numerical replay. This validates the declared local experiment, not equality with the original paper.

RouterBench. At 0.5 of the reference budget, PILOT delivered 0.676 reward per requested query with 100.0% coverage. The highest mean at this budget was LinUCB (0.706). These are descriptive results on fixed streams.

MMLU subset. At 0.5 of the reference budget, PILOT delivered 0.667 reward per requested query with 100.0% coverage. The highest mean at this budget was ExploreOnly (0.690). These are descriptive results on fixed streams.

Accounting: 12 of 1,098 conditions recorded actual benchmark spend above the estimated budget. The selector enforces estimated affordability; actual recorded costs are reported separately. These costs are historical benchmark units, not charges incurred by this experiment.

Method	RouterBench utility	RB coverage	MMLU utility	MMLU coverage
PILOT	0.6764	100.0%	0.6675	100.0%
LinUCB	0.7056	100.0%	0.6878	100.0%
EpochGreedy	0.6973	100.0%	0.6818	100.0%
ExploreOnly	0.7031	100.0%	0.6899	100.0%
Random	0.5207	100.0%	0.4667	100.0%
Fixed WizardLM 13B	0.4307	100.0%	0.4369	100.0%
Fixed Claude Instant	0.5932	100.0%	0.5835	100.0%
Fixed Claude v1	0.0091	1.4%	0.0212	3.0%
Fixed GPT-3.5	0.6092	100.0%	0.6212	100.0%
Fixed GPT-4	0.0064	0.9%	0.0165	2.3%
Fixed Code Llama 34B	0.1986	100.0%	0.0031	100.0%
Fixed Llama 2 70B	0.3274	100.0%	0.0282	100.0%
Fixed Yi 34B	0.6444	100.0%	0.6455	100.0%

Table: mean across three seeds, full learning, budget fraction 0.5. Full seed/budget tables accompany this report.

Design, controls and interpretation

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Research basis. Panda et al. formulate adaptive LLM routing as a contextual bandit, initialize it using human preferences, and combine routing with an online budget policy. Our implementation follows this structure with explicitly documented local substitutions. Reference: Adaptive LLM Routing under Budget Constraints, Findings of EMNLP 2025, DOI 10.18653/v1/2025.findings-emnlp.1301.

Data and representation. Pinned RouterBench 0-shot data: 1,000 tuning / 32,270 learning / 3,227 deployment queries. MMLU retains the same frozen phase order after task-prefix filtering: 398 / 12,369 / 1,275. MMLU is nested in RouterBench. Arena preparation supplies 1,245 training and 313 holdout preference pairs. Local BGE-M3 encodes all inputs at a 2,048-token cap; no routing input is truncated, and 13 preference inputs are truncated.

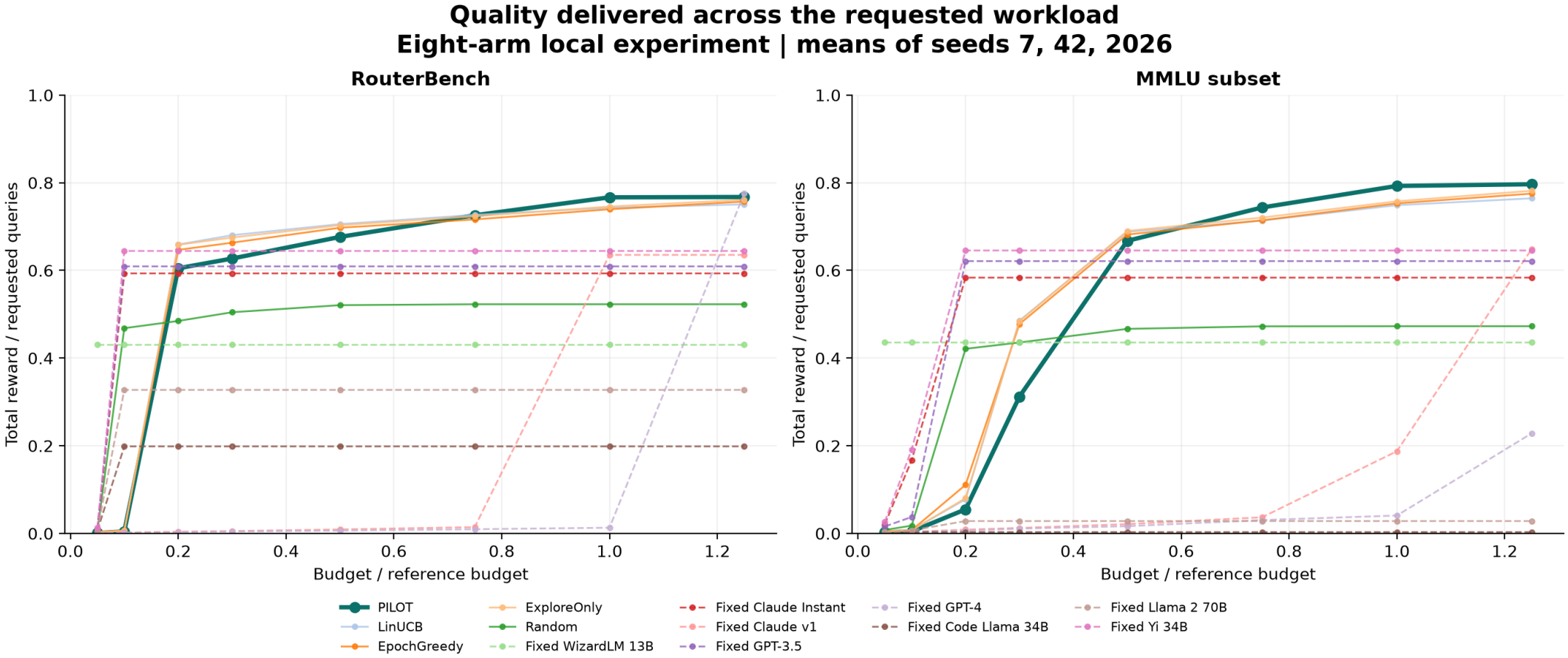
Training and selection. Each seed (7, 42, 2026) retrains a 64D projection/prior for 30/40 epochs, learning rate 0.03 and margin 0.2. Independent holdout checks determine finite initialization coefficients for every arm. Tuning selects PILOT alpha=1, LinUCB alpha=5 and EpochGreedy window=10 in both domains. Tuning states never initialize routing learning. Only chosen-arm feedback updates learning; deployment uses a frozen checkpoint.

Common comparison. Every method uses identical query order, model pool, tuning-only length estimates and absolute budget per condition. Budget fractions are 0.05, 0.1, 0.2, 0.3, 0.5, 0.75, 1 and 1.25 times the domain reference budget. Reference budgets are 10.23165528 for RouterBench and 1.7940659547738693 for MMLU. Bin size is 100. Linear predictions are unclipped; random/fixed scores are declared utility conventions, not calibrated probabilities.

What the metrics mean. Service utility is total observed reward divided by all requested queries; unserved service utility is zero. Coverage and served-query mean are separate. Learning regret uses the best stored reward among the eight admitted arms as a retrospective oracle; the policy never receives that oracle. Seed mean/min/max describes these three runs, not a confidence interval or significance test.

Quality delivered under each budget

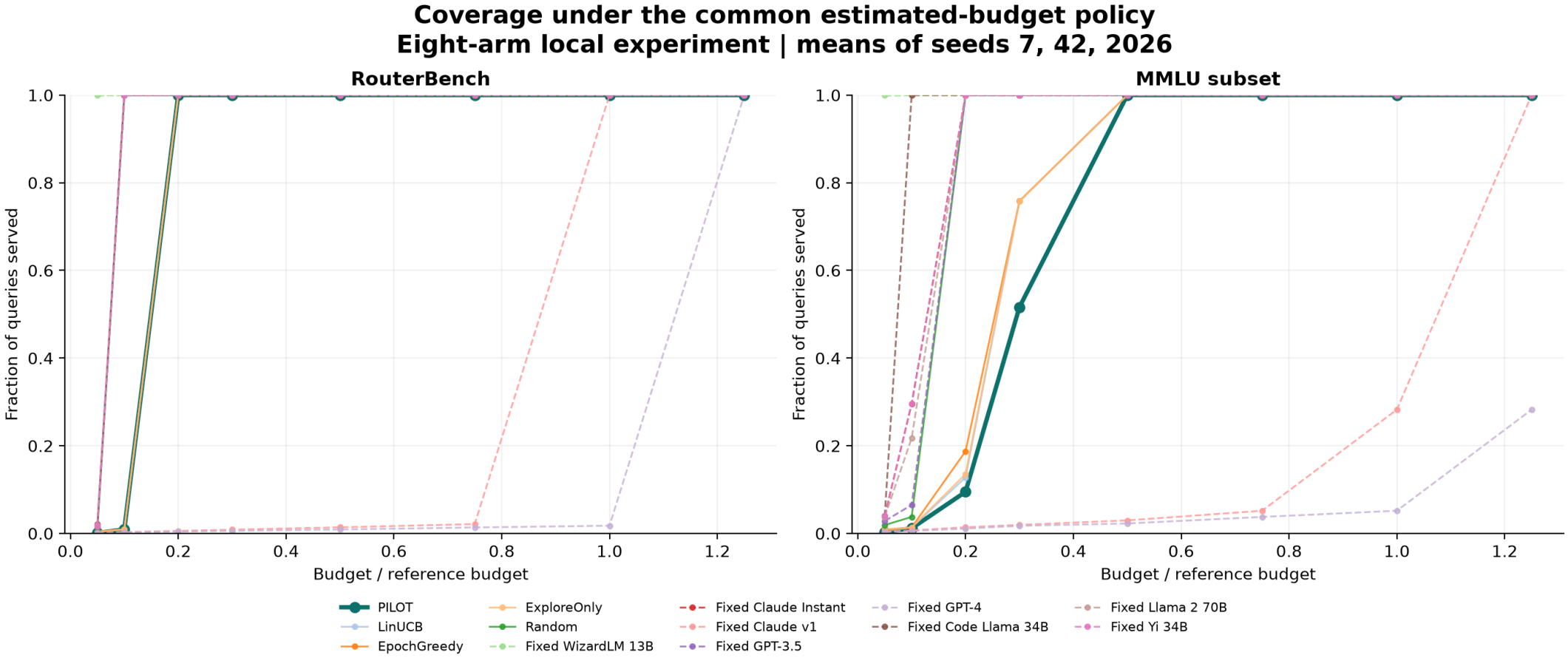
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All 13 methods and all eight budget fractions. Lines show the mean over the three fixed seeds. Unserved queries contribute zero service utility. Full per-seed results remain in the CSV tables.

Budget exhaustion and served coverage

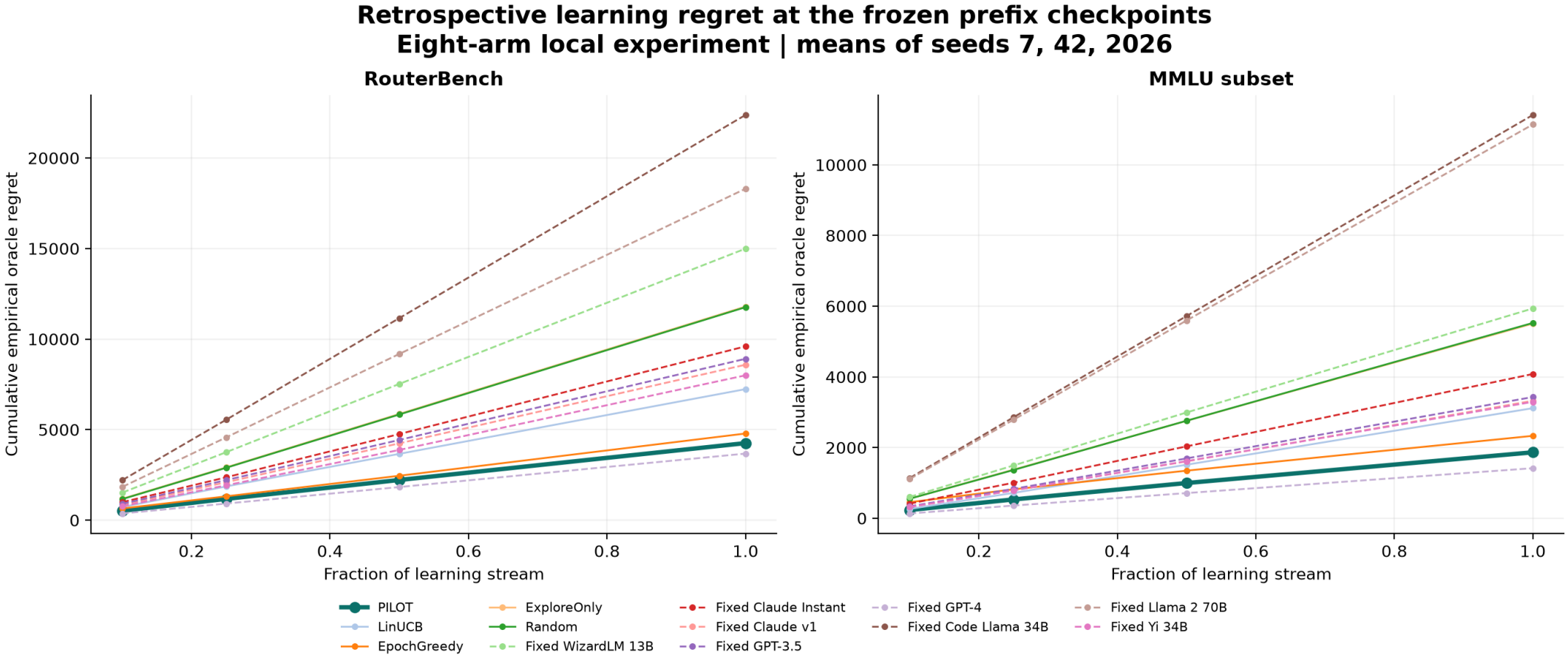
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Coverage is essential to interpreting reward on served queries. The common policy terminates a stream when neither its threshold rule nor its affordability fallback admits a permitted model. Fixed policies may use only their named arm.

Learning feedback and empirical regret

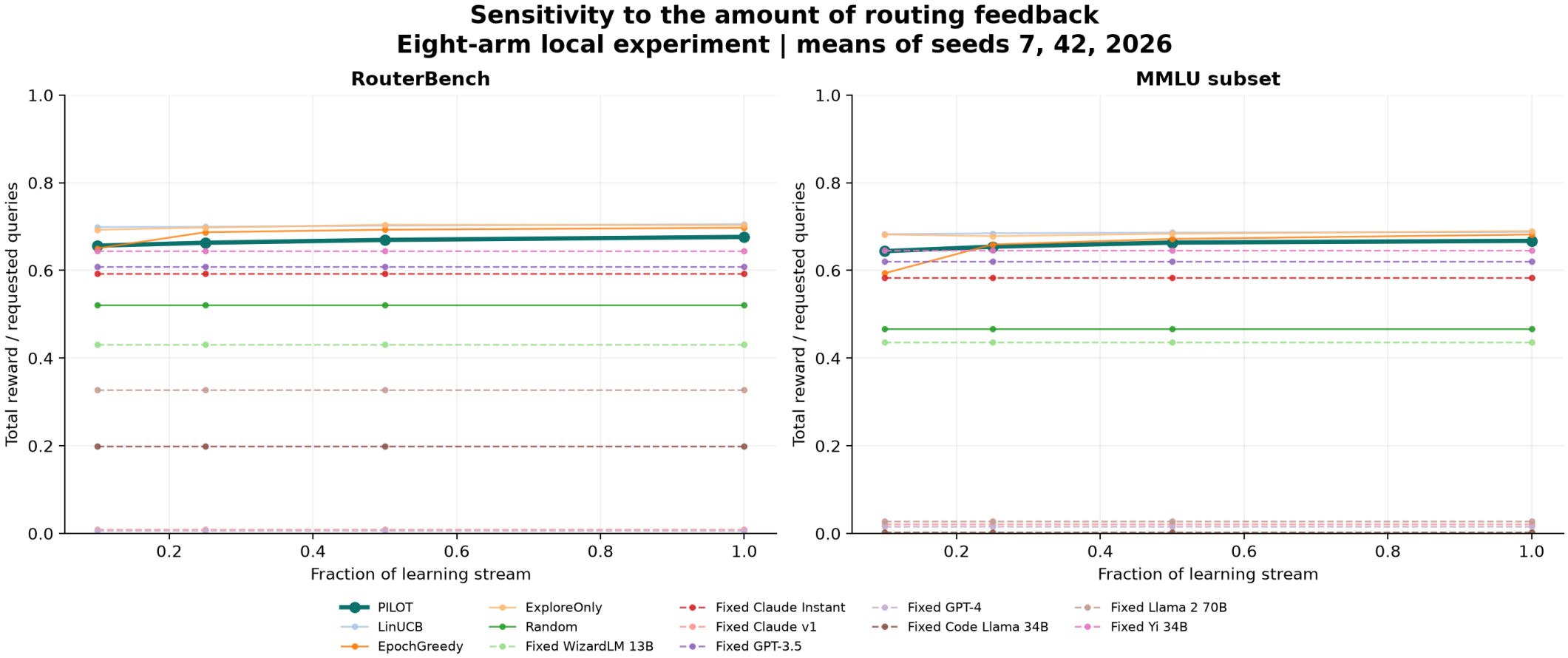
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Snapshots at 10%, 25%, 50% and 100% are nested prefixes of one uninterrupted stream. The oracle is retrospective and unconstrained by deployment budgets. Different stream lengths make the two domains separate descriptive panels.

Sensitivity to routing-learning data

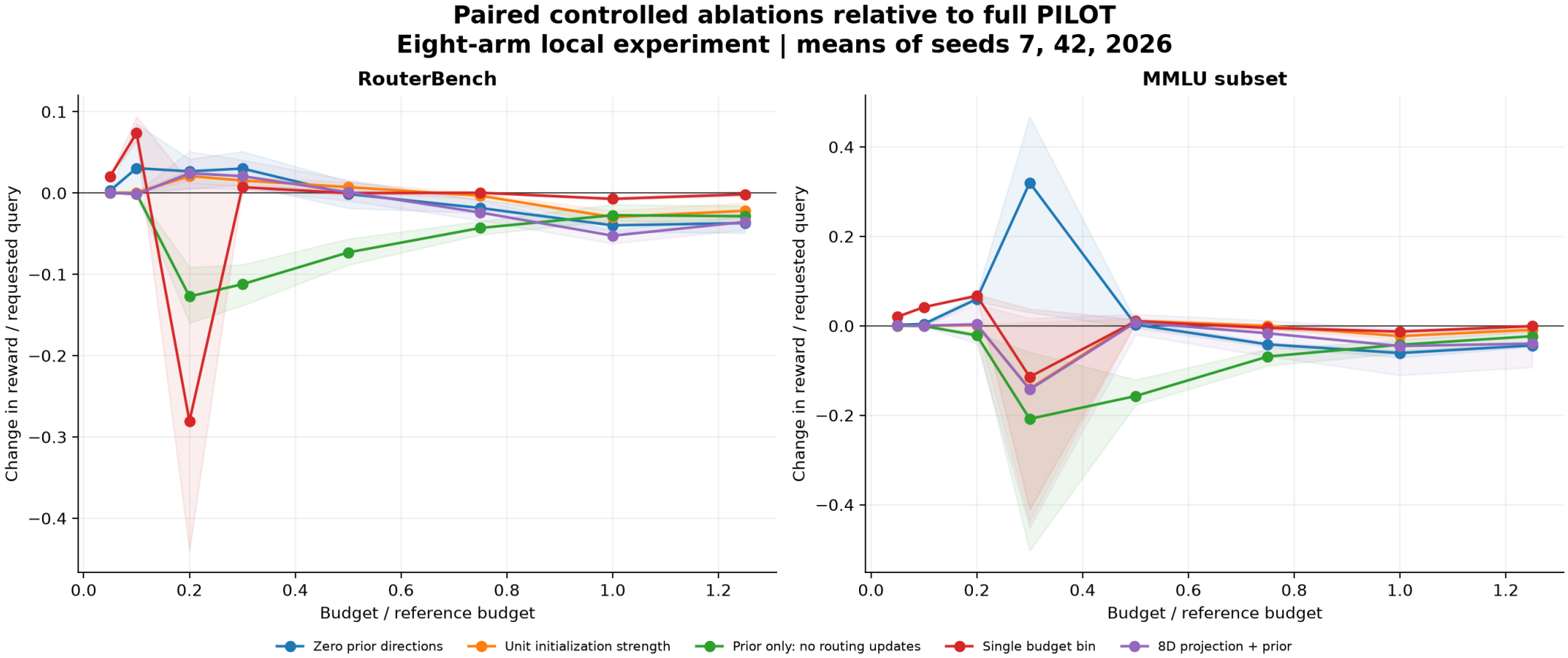
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All four prefix sizes use the same deployment stream and budget fraction 0.5. Preference prior, seed, cost estimates and policy parameters stay fixed within a replicate; no tuning occurs on deployment outcomes.

Controlled changes relative to full PILOT

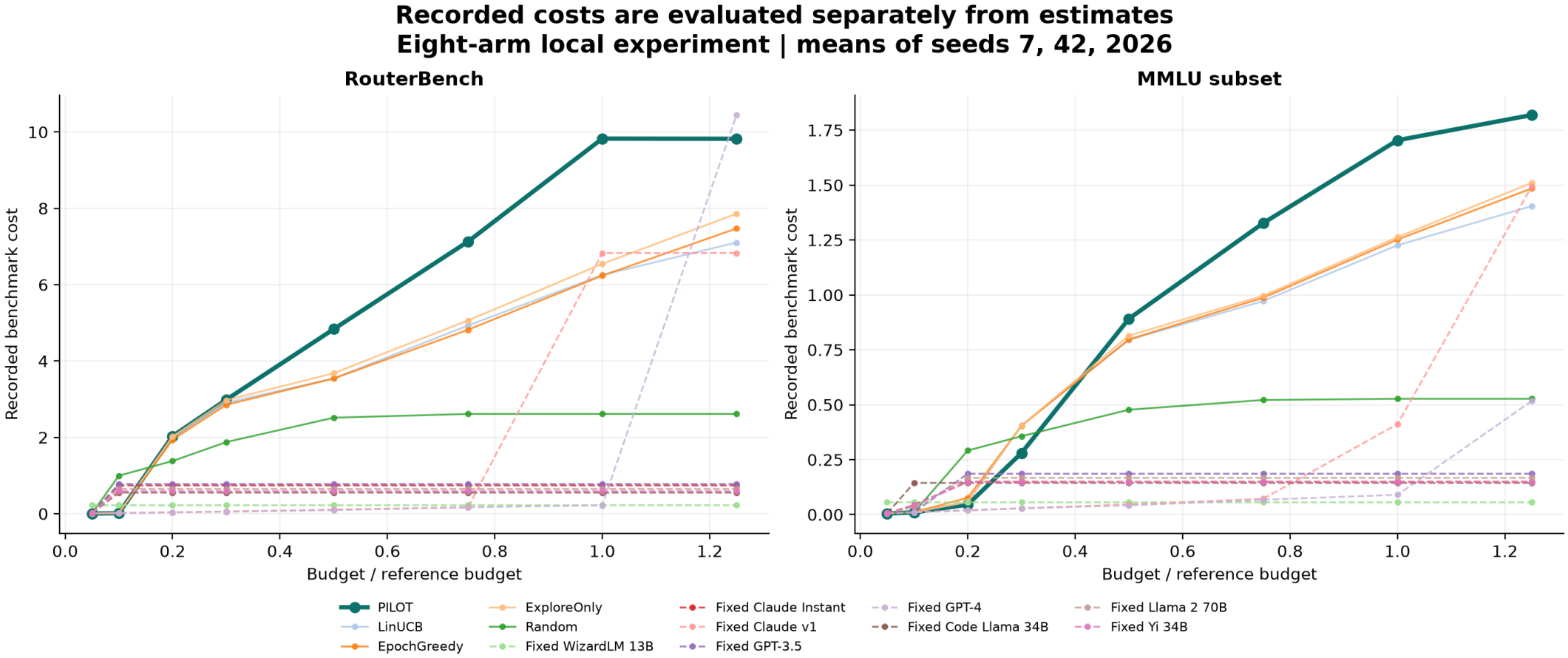
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All five declared variants are reported, including unfavorable or identical outcomes. Bands are observed min/max paired seed deltas, not confidence intervals. The 8D variant retrains the same 30/40-epoch recipe; it is not an embedding-provider comparison. Zero prior directions retain the learned projection and coefficients.

Recorded cost versus declared budgets

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These are recorded benchmark costs, separately accumulated after routing decisions. Output-length estimates come only from tuning. An estimated spending bound does not guarantee a bound on recorded actual spending; per-run excess is included in the tables.

Limits, evidence and reproduction

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- The model pool has eight arms. Claude-v2, Mistral-7B-chat and Mixtral-8x7B-chat remain excluded because their provider checkpoint bridges are unresolved. Canonical name identity alone does not establish serving equivalence.

- The original paper release/shot selection, 86 local task labels versus 64 reported tasks, and exact Arena preparation remain unresolved. This study uses a declared 313-pair holdout rather than claiming the paper's 500-example evaluation.

- BGE-M3, the 2,048-token cap, 64D/8D projections, initialization/training recipe and baseline deployment score conventions are explicit local adaptations. An 8D sensitivity result is not evidence that BGE-M3 matches the paper's embedding provider.

- Historical price inputs use a documented factor-of-ten dataset/source calibration and a shared token-count approximation. Triplet cost ordering assumes a 1:1 input/output reference mix. Actual cost overruns and score-scale differences must remain visible.

- Three seeds and nested MMLU do not establish population uncertainty or independent dataset replication. The complete grid is reported; individual favorable configurations are not treated as general superiority.

Evidence: aggregate/manifest.json pins every completed run and all four CSV tables. verification/ contains independent action, state, cost and metric replay for every condition group. figures/figure-data.json records the exact numeric series behind every figure. Raw prompts and generated vectors stay out of Git.

Reproduction: docs/full-experiments.md lists the command sequence and immutable resume behavior. Use configs/frozen_experiment_protocol.json and configs/research_prior_index.json without changing pins. Run one job/worker at a time with at least 8 GiB available RAM. No provider generation API is called.

Reference: Panda et al. (2025), Adaptive LLM Routing under Budget Constraints. Dataset and cost-source identities are recorded in the frozen protocol and docs/source-cost-evidence-2026-10-29.md.